

ME2 — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

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October 2026

CRNN · 718,932 params

20,611 clips

Trained on A100

Validated on RPi 4

● **MODEL**

Architecture	CRNN (Conv x2 → Bi-LSTM x2)
Parameters	718,932
Weights	2.743 MB fp32 · 702 KB int8
Input feature	MFCC(40) · 100 frames
Output classes	20 (19 intents + REJECT)
Slotted intents	6 (timer, alarm, temp, bright, color, reminder)

● **DATASET**

Source	HF · airimonda/ai231-me2-voice-commands
Clips / hours	20,611 clips · 16 kHz mono
Speakers	260 train / 115 test / 5 holdout
Voice mix	≈85% synthetic + ≈15% real (train)
Labels	19 intents · 93 phrases · 6 slotted
Split (speaker-disjoint)	15,716 / 4,693 / 202

● **TRAINING ON THE A100**

Cluster	1× A100-SXM4-40GB (40 GB HBM2e)
Objective	Cross-entropy · 20-way classification
Optimizer	Adam (lr 1e-3, cosine) · seed 42
Batch / patience	256 · patience 10 (max 100 ep)
Convergence	best ep 29 · early stop ep 39
Wall-clock	≈25 s · peak VRAM < 1 GB

● **VALIDATION ON THE RASPBERRY PI 4**

Intent accuracy	79.12% (3713/4693)
Task completion	79.12% · Cohen's κ 0.7765
Macro F1	0.7672
Out-of-scope recall	74.23% · false-reject 7.17%
Inference latency	p50 1.995 ms · p95 2.731 ms
Wake word (openWakeWord)	p50 2.23 ms / 80 ms chunk
Runtime	onnxruntime · int8 · CPU

PIPELINE — MIC TO ACTUATOR

Mic
16 kHz mono

→

Wake word
openWakeWord
"hey jarvis"

→

Features
MFCC(40)
100 frames

→

Encoder
Conv x2 →
Bi-LSTM x2

→

Intent head
Linear → 20
+ slot values

→

Actuator
Raspberry Pi 4
(4 GB)

What the reviewer will look for

Status of each deliverable for this submission

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CRNN · 20,611 clips · A100 → RPi 4

REVIEWER CHECKLIST		
1	Repo public, one-command reproduction	done · launch_web.sh
2	Dataset licensed and citable (DOI)	done · HF dataset card
3	Training logs + final checkpoint committed	done · crnn_a100_best.pt
4	Pi 4 latency reproduced by the posted script	pending on-Pi re-run · reported p50 1.995 ms
5	Held-out test set, unseen speakers	done · 115 test speakers
6	Baseline of comparable size compared	done · 5-model benchmark

● REFERENCES & ARTEFACTS

GitHub repository	github.com/alyannalopez/MLOps-Exercises
Dataset location	HF · airimonda/ai231-me2-voice-commands (access terms on card)
A100 cluster	1× A100-SXM4-40GB · CUDA 12.4 · torch 2.6.0 · seed 42
Model weights	crnn_a100_best.pt · crnn_float.onnx · crnn_int8.onnx
Wake word	openWakeWord "hey jarvis" · MIT · dscripka/openWakeWord

Best result: A100 retrain (no weight decay), test accuracy 0.7912 on 4,693 held-out clips (3,713 correct). Local-MPS reference run 0.7765; A100 + weight-decay 0.7699. All figures taken from report/report.md, models/*_result.json and report/wakeword_info.json.